



SniperSheet

AI-Powered Excel Formula Engine

Natural Language → Precise Excel Formulas in Seconds

35+

Formula Patterns

4

Smart Tabs

100%

Free AI

3

AI Models

2

Languages

Smart Hub

Commands

Cell Dimensions

Advanced Tools

Local Fallback

Free Forever

What is SniperSheet?

Complete product overview and key capabilities

SniperSheet is a professional Microsoft Excel task pane add-in that transforms natural language descriptions into precise, ready-to-use Excel formulas. Powered by Groq AI (Llama 3.3 70B — completely free), it supports both English and Arabic input, runs on a US-based Replit server so users worldwide can access it without VPN, and falls back instantly to a local formula engine when offline.

Groq AI Engine

Llama 3.3 70B model
100% free, no credit

Bilingual Support

English & Arabic input
Full RTL layout

Proxy Architecture

US Replit server relays
all AI calls — no VPN

Local Fallback

35+ instant patterns
Zero latency fallback
Works completely offline

Office.js Integration

ReadWriteDocument
permission level
Excel Ribbon button

Responsive Design

Native scroll — 400px
task pane optimized
Mobile-friendly UX

Just describe your calculation in plain English or Arabic — SniperSheet generates the exact Excel formula with explanation & confidence score



Smart Hub — AI Formula Engine

Tab 1 · Natural language → Excel formula via Groq AI



Smart Hub

AI Powered

The core feature of SniperSheet. Type any description of what you want to calculate — in English or Arabic — and click "Smart Analysis". The AI engine processes your request through Groq AI (Llama 3.3 70B) and returns the exact Excel formula, a full explanation, a

- AI-powered formula generation (Llama 3.3 70B)
- Confidence scoring system: 0–100%
- Formula type auto-detection (IF, VLOOKUP, SUM...)
- Word Radar: detects typos & ambiguous terms

confidence score (0–100%), and optional conditional

- Style hints: background color, bold, italic

Commands — Arabic Formula Aliases

Tab 2 · Arabic & English command names for instant calculations

- Full history log with timestamps



Commands

Arabic Aliases

Type the function name in English or Arabic (e.g., "جمع" for SUM, "ضرب" for MULTIPLY) along with a cell range, and the system instantly calculates the result and generates the corresponding Excel formula. Includes a searchable command reference table and full execution history.

- Example prompts library (EN + AR)

- Arabic aliases: جمع, ضرب, مكافئة متوسط
- SUM, MULTIPLY, AVERAGE, MAX, MIN
- COUNT, BONUS (15%), TAX (15%), PERCENTAGE
- Real-time calculation & formula display
- Searchable command reference table
- Execution history with timestamps

Cell Dimensions

Tab 3 · Calculate optimal cell width & height for any text

Cell Dimensions

Arabic-Aware

Calculates the optimal width and height for an Excel cell based on its content, accounting for Arabic character width, font size, bold, italic, and padding. Supports single cell and batch processing modes.

- Arabic text character width calculation
- Font size & bold/italic awareness
- Single cell & batch processing
- Pixel and Excel unit output
- Recommended padding included
- One-click copy to clipboard

Advanced Tools

Tab 4 · Three professional productivity utilities

Empty Field Radar

Scan CSV or tab-separated data for missing values. Paste your data, click scan, and get a visual color-coded grid plus a precise list of every empty cell position.

- Paste CSV / tab-separated data
- Visual grid: filled=green, empty=red
- Exact cell coordinate list
- Header row auto-detection
- Filled vs empty count summary

Smart Print-Fit

Calculate optimal column widths and font sizes to fit your spreadsheet onto a specific paper size, eliminating trial-and-error print formatting.

- A4 & A3 paper support
- Portrait & landscape orientation
- Excel column width units output
- Recommended font size per row
- Print ready in 3 clicks

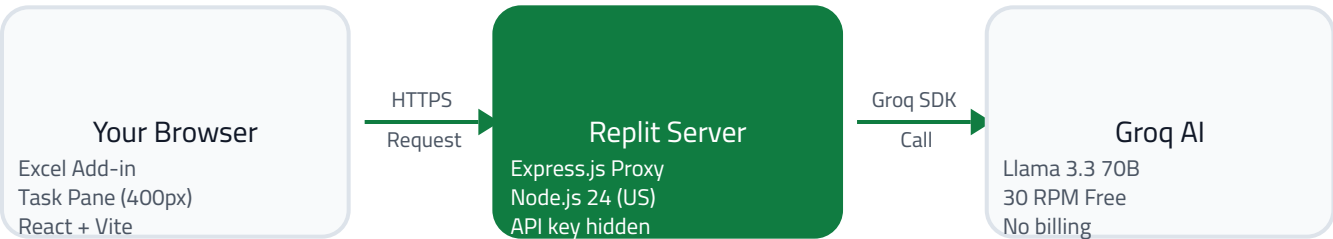
Professional Report View

Transform raw pasted data into a styled, professional HTML table preview with alternating row colors, then send directly to the browser print dialog.

- Styled HTML table from raw data
- Alternating row color scheme
- Sortable column headers
- Auto header row detection
- One-click browser print dialog

AI Proxy Architecture

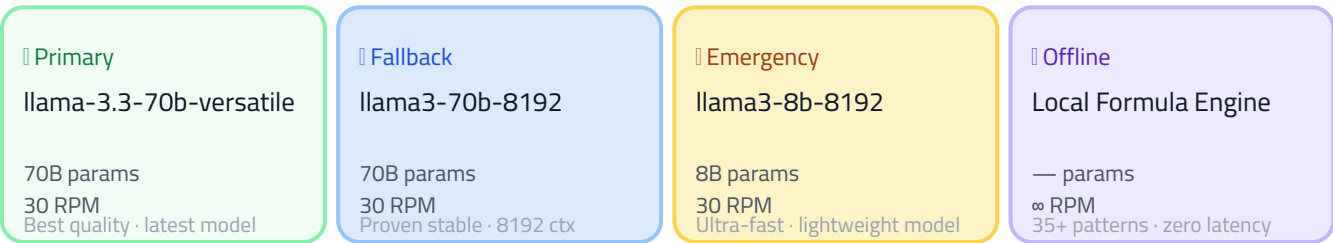
How SniperSheet connects to AI without exposing API keys to users



Security: The GROQ_API_KEY is stored as a Replit secret server-side. Users in Iraq (or anywhere) connect only to the HTTPS Replit endpoint — they never touch the Groq API directly, so no VPN is needed.

AI Model Cascade

3 Groq models + local fallback — automatic retry on failure



window

Technical Specifications

Complete add-in configuration and capabilities

Office.js Version	Frontend
Manifest 1.1	React 18 + Vite + TypeScript
Add-in Type	Backend
Task Pane	Express.js + Node.js 24
Task Pane Width	AI SDK
400px	Groq SDK (Official)
Permission Level	Package Manager
ReadWriteDocument	pnpm Monorepo Workspace
Default Locale	Deployment
ar-SA (Arabic)	Replit Autoscale
Ribbon Tab	Styling
Custom "SniperSheet" tab	Tailwind CSS + shadcn/ui

Real Formula Examples

Generated by the Smart Hub AI engine from natural language

Formula Generated	Natural Language Input	Category
=IF(A1>40, (A1-40)*1.5, 0)	Calculate overtime: if hours > 40, multiply excess by 1.5	Conditional
=IFS(B1>=90, "A+", B1>=75, "B", ...)	Grade student: A+ above 90, B above 75, else C	Conditional
=XLOOKUP(D1, A:A, B:B, "Not found")	Find value in column A, return matching value from	Lookup
=SUMIF(C:C, ">1000", D:D)	Sum all sales where amount is greater than 1000	Statistical
=PMT(0.05/12, 360, A1)	Monthly payment for 30-year loan at 5% annual	Financial
=DATEDIF(A1, TODAY(), "Y")	Calculate person age in years from their birth date in A1	Date
=RANK(A1, A:A, 0)	Rank employee by highest sales figure in the column	Statistical
=IF(A1>10000, A1*1.15, A1)	Add 15% bonus to salary if sales exceed 10,000	Conditional

Installation Guide

4 simple steps to load SniperSheet in Microsoft Excel

1

Download manifest.xml

Visit your Replit deployment URL and download the add-in manifest file from the /api/addin/manifest.xml endpoint

2

Open Excel ▯ Insert ▯ Add-ins

In Microsoft Excel, click Insert in the ribbon, then select Add-ins ▯ My Add-ins ▯ Manage My Add-ins.

3

Upload manifest.xml

Click "Upload My Add-in", select the manifest.xml file you downloaded. A new "SniperSheet" tab will appear in the ribbon

4

Open Sniper Hub

Click the "SniperSheet" ribbon tab, then click "Open Sniper Hub". The task pane opens on the right side — you're ready!



SniperSheet

AI-Powered Excel Formula Engine · Official English Edition

Mustafa Alsahlany

Developer & Designer of SniperSheet Excel Add-in
Built with React, Express.js, TypeScript, and Groq AI on Replit

- ▮ Replit — Active Deployment
- ▮ Groq AI — Free Tier
- ▮ Version 1.0.0
- ▮ April 2026

▮ Technology Stack

- React 18 + Vite (Frontend)
- Express.js + Node.js 24 (Backend)
- Groq SDK — Llama 3.3 70B (AI)
- Tailwind CSS + shadcn/ui (Design)
- TypeScript — Full Stack
- pnpm Monorepo Workspace

▮ Add-in Specifications

- Office.js Manifest 1.1
- Task Pane · 400px width
- Permission: ReadWriteDocument
- Custom Excel Ribbon tab
- Default Locale: ar-SA
- HTTPS required (Replit hosted)

